

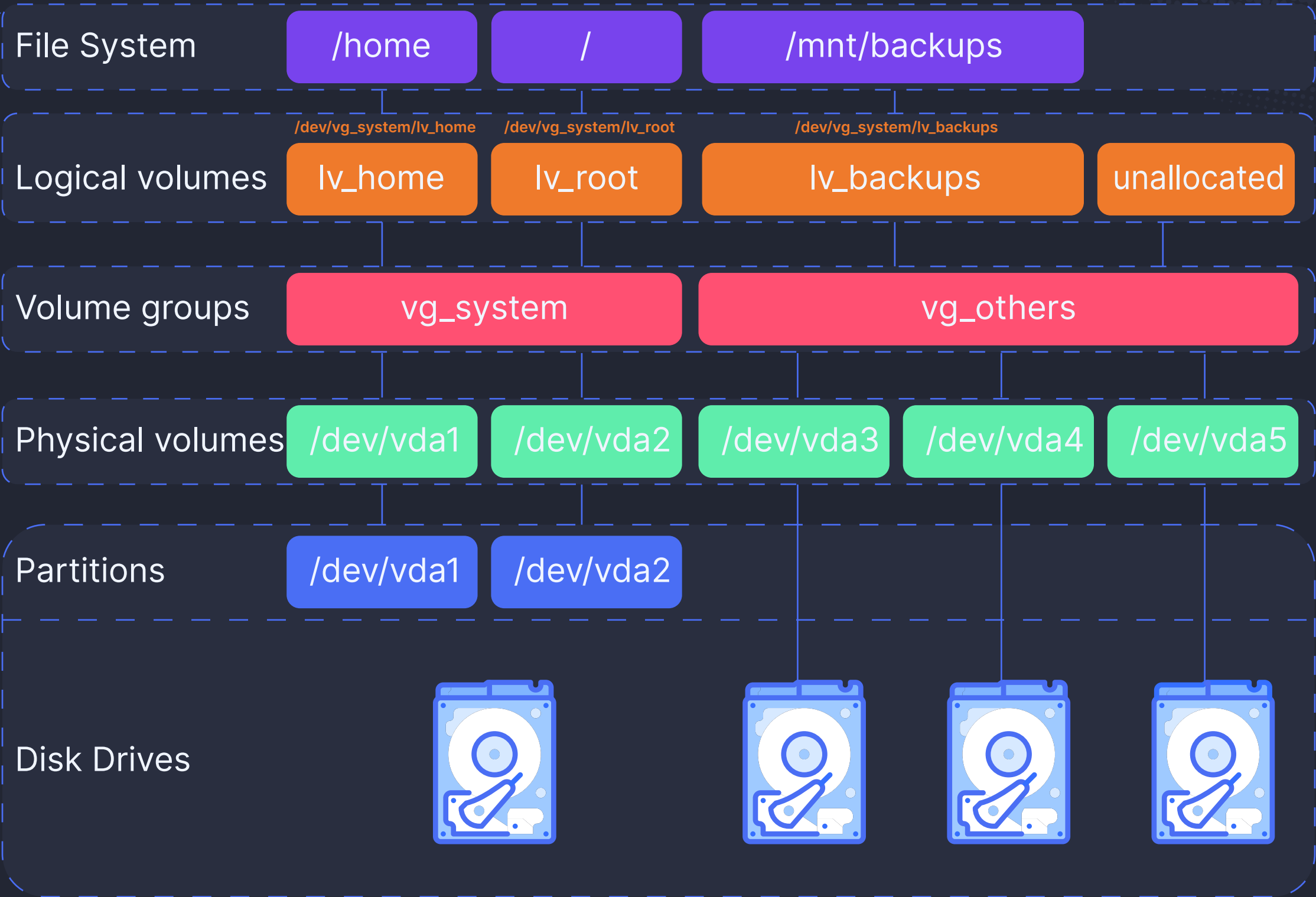
# Linux Logical Volume Manager

The filesystem sits on top of the logical volume and it will be formatted to a specific fs type (vfat, xfs, ext4) and mounted anywhere on the system

A Logical Volume sits inside a Volume Group and it's what is assigned to a file system.

A Volume Group can have multiple physical Volumes

Physical volumes are directly related to hard drives or partitions



## Volume Group Management Commands

| Command        | Description                                     |
|----------------|-------------------------------------------------|
| vgcreate       | Create a volume group                           |
| vgscan         | Search for all volume groups                    |
| vgdisplay, vgs | Display information about vgs                   |
| vgextend       | Add physical volumes to a vg                    |
| vgremove       | Remove volume group(s)                          |
| vgrename       | Rename a volume group                           |
| vgchange       | Change volume group attributes                  |
| vgck           | Check the consistency of vgs                    |
| vgmerge        | Merge volume groups                             |
| vgsplit        | Move pvs into a new or existing vg              |
| vgcfgbackup    | Backup vg configuration(s)                      |
| vgcfgrestore   | Restore vg configuration                        |
| vgconvert      | Convert vg metadata format                      |
| vgexport       | Unregister vgs from the system                  |
| vgimport       | Register exported vg with system                |
| vgimportclone  | Import a vg from cloned pvs                     |
| vgmknodes      | Create the special files for vg devices in /dev |

## Physical Volume Management Commands

| Command     | Description                                                 |
|-------------|-------------------------------------------------------------|
| pvcreate    | Initialize a disk or partition for use as a physical volume |
| pvscan      | Scan all disks for pvs                                      |
| pvdisk, pvs | Display information about pvs                               |
| pvresize    | resize a physical volume                                    |
| pvmove      | Move extents from one physical volume to another            |
| pvck        | Check metadata on pvs                                       |
| pvremove    | Remove LVM label(s) from pvs                                |
| pvchange    | Change attributes of physical volumes                       |

## Logical Volume Management Commands

| Command            | Description                                      |
|--------------------|--------------------------------------------------|
| lvcreate           | Create a logical volume                          |
| lvscan             | Scan (all disks) for lvs                         |
| lvdisplay, lvs     | Display info about lvs                           |
| lvextend           | Extend size of a lv                              |
| lvremove           | Remove a logical volume                          |
| lvrename           | Rename a logical volume                          |
| lvchange           | Change attributes of a lv                        |
| lvreduce, lvresize | Reduce and resize the size of a logical volume   |
| lvconvert          | Convert a logical volume from linear to mirrored |

## Examples

### Create Pyhical Volumes

```
$ sudo pvcreate /dev/vda1 /dev/vda2 /dev/vda3 /dev/vda3 /dev/vda5
```

### Create Volume groups

```
$ sudo vgcreate vg_system /dev/vda1 /dev/vda2
$ sudo vgcreate vg_others /dev/vda3 /dev/vda3 /dev/vda5
```

### Create Logical Volumes

```
$ sudo lvcreate -L 20GB -n lv_home vg_system
$ sudo lvcreate -L 35GB -n lv_root vg_system
$ sudo lvcreate -L 70GB -n lv_backups vg_others
```